

```
import pandas as pd

import matplotlib.pyplot as plt


# -----

# 1. Create DataFrames

# -----

students = pd.DataFrame({
    "ID": [1, 2, 3, 4, 5],
    "Name": ["Arun", "Bala", "Cathy", "David", "Eva"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE"]
})

marks = pd.DataFrame({
    "ID": [1, 2, 3, 4, 5],
    "Marks": [85, 65, 90, 70, 78]
})


# -----

# 2. Combining

# -----

df = pd.merge(students, marks, on="ID")

print("Combined Data")

print(df)
```

```
# -----
```

```
# 3. Indexing
```

```
# -----
```

```
print("\nFirst Student")
```

```
print(df.iloc[0])
```

```
print("\nName and Marks")
```

```
print(df[["Name", "Marks"]])
```

```
# -----
```

```
# 4. Filtering
```

```
# -----
```

```
print("\nStudents with Marks > 75")
```

```
print(df[df["Marks"] > 75])
```

```
# -----
```

```
# 5. Grouping
```

```
# -----
```

```
print("\nDepartment Average")
```

```
print(df.groupby("Department")["Marks"].mean())
```

```
# -----
```

6. Sorting

```
sorted_df = df.sort_values(  
    "Marks",  
    ascending=False  
)
```

```
print("\nSorted Marks")
```

```
print(sorted_df)
```

7. File I/O

```
df.to_csv(  
    "student_analysis.csv",  
    index=False  
)
```

```
print("\nFile Saved Successfully")
```

8. Plotting

```

df.plot(
    x="Name",
    y="Marks",
    kind="bar"
)

plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")

plt.show()

```

OUTPUT:

Combined Data

ID	Name	Department	Marks
01	Arun	CSE	85
12	Bala	ECE	65
23	Cathy	CSE	90
34	David	ECE	70
45	Eva	CSE	78

First Student

ID 1

Name Arun

Department CSE

Marks 85

Name: 0, dtype: object

Name and Marks

	Name	Marks
0	Arun	85
1	Bala	65
2	Cathy	90
3	David	70
4	Eva	78

Students with Marks > 75

	ID	Name	Department	Marks
0	1	Arun	CSE	85
2	3	Cathy	CSE	90
4	5	Eva	CSE	78

Department Average

Department

CSE 84.333333

ECE 67.500000

Name: Marks, dtype: float64

Sorted Mark

	ID	Name	Department	Marks
23	Cathy	CSE	90	
01	Arun	CSE	85	

45	Eva	CSE	78
34	David	ECE	70
12	Bala	ECE	65

File Saved Successfully

File Saved Successfully

